

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Computer Science and Engineering(Artificial Intelligence and Machine Learning)) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering(Artificial Intelligence and Machine Learning)) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTAIPE603D - WEB DEVELOPMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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1. What is Node.js primarily used for?
a) Styling web pages b) Running JavaScript on the server
c) Designing databases d) Writing HTML code
2. Which of the following is a component of the Struts framework?
a) ServletContainer b) Action class c) JSPBean d) SpringBoot
3. Which of the following is used to validate the structure of an XML document?
a) HTML b) JavaScript c) DTD d) JSON
4. Which DOM method is used to get an element by its ID?
a) getElementByClass() b) getElementById()
c) querySelectorAll() d) findById()
5. Which tag is used to create a hyperlink in HTML?
a) <a> b) <link> c) <href> d) <url>
6. Which lifecycle method is called when a servlet is being removed from memory?
a) init() b) doGet() c) service() d) destroy()
7. Which component in Struts is responsible for handling the client request?
a) View b) Action class c) Interceptor d) Controller Servlet
8. In XML, a declaration is used at the beginning of every well-formed document.
a) `<?xml version="1.0"?>` b) `<!DOCTYPE>`
c) `<!-- comment -->` d) `<xml>`
9. Which of the following is a valid way to define a function in JavaScript?
a) `function = myFunc()` b) `function myFunc() { }`
c) `function:myFunc()` d) `def myFunc()`
10. Which method is called only once during the lifecycle of a servlet?
a) doGet() b) doPost() c) init() d) service()
11. Which of the following is used to describe a web service?
a) UDDI b) WSDL c) XML d) SOAPUI

- 12 PHP global variable retrieves data sent with a form using POST method
a) \$_FORM b) \$_POST c) \$_GET d) \$_REQUEST

Q2. Solve the following.

- A) Describe the structure of an HTTP request message. Create a simple HTML page with a heading, a paragraph, and an image. 6
- B) Design and implement a complete HTML form for user registration that includes the appropriate fields: Full Name, Email Address, Password, Gender, Date of Birth, Phone Number, Address, City, Interests, and Submit and reset Buttons. 6

Q3. Solve the following.

- A) Distinguish between primitive and reference data types in JavaScript. Explain the difference between var, let, and const in JavaScript with a suitable example. 6
- B) Write a JavaScript function to find the largest number among three numbers entered by the user and return the result. 6

Q4. Solve Any Two of the following.

- A) Describe the servlet architecture and how it processes client requests. Discuss how parameter data is received and processed in servlets with example. 6
- B) Describe the purpose of XML namespaces with a suitable example. Define DTD, discuss the type of DTD, give an example of any one of its types, and explain the role of DTD in XML validation. 6
- C) What is the purpose of *web.xml* in a servlet project? Name two XML schema definition types. Write an XML document to represent a student record that includes: student ID (as an attribute), name, age, gender, course, year of enrollment, email, and address (with nested elements for street, city, and postal code). Ensure the XML is well-formed and structured. 6

Q5. Solve Any Two of the following.

- A) Explain the relationship between JSP and Servlets. Describe the process of running a JSP application. 6
- B) Write a client program to consume a Java web service using HTTP to call the following webpage: "http://localhost:8090/MyWebService/ApplicationForm" and return plain text or JSON data. 6
- C) Implement form validation in a Struts web application using the built-in Struts Validation Framework. Create a user registration form with fields: username, password, email, and age. Apply the validation rules and ensure meaningful error messages are displayed when validation fails. Use *validation.xml* or annotations for defining validation rules. 6

Q6. Solve Any Two of the following.

- A) Explain the basic syntax and structure of a PHP script. Describe how PHP handles form data using *\$_POST* and *\$_GET*. 6
- B) Explain the steps in establishing a connection to a MySQL database using PHP, and illustrate with appropriate syntax. 6
- C) What is the *.NET* framework? Define a web service in *ASP.NET*. Write a simple C# program to demonstrate control flow using *if-else* and loops. 6

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